Exchange Rate Regimes in Production Networks

A Small Open Economy Extension of Rubbo (2024)

Sugarkhuu Radnaa

Bonn Graduate School of Economics

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Building on: Rubbo (2024) *Networks, Phillips Curves, and Monetary Policy*

Outline

Motivation: Networks Meet the Exchange Rate

Two literatures that rarely talk to each other:

1. **Production-network Phillips curves** (Rubbo 2024; La'O & Tahbaz-Salehi 2022): with input-output linkages, sector-level price stickiness generates a genuine inflation–output tradeoff for *almost every* price index — except one. A unique "divine coincidence" (DC) index, weighting sectors by Domar-scaled stickiness, restores the tradeoff-free policy prescription. All of this is derived in a **closed economy**.
2. **Small open economy NK models** (Galí & Monacelli 2005 and successors): the exchange rate is a cost-push shifter and a policy instrument, but the production side is a single aggregate sector — no network structure, no heterogeneity in who is exposed to FX shocks.

This paper puts them together. Once imported inputs enter a multi-sector network, the exchange rate becomes a *second* cost-push channel that hits sectors asymmetrically depending on their import centrality and their position in the domestic supply chain. Does the DC index — or any analog — survive, and which FX regime is actually optimal once network structure is taken seriously?

Why This Matters: The Policy Question

Motivating case: a small oil-exporting economy. Resource sector is upstream, import-intensive (machinery) and export-heavy; Manufacturing sits in the middle; Services is downstream, domestically oriented, and dominates final consumption.

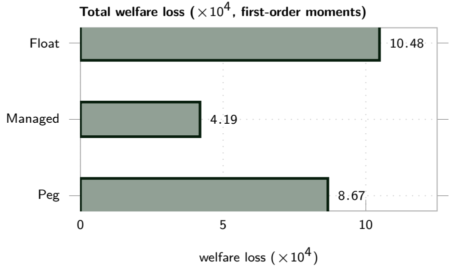
The question a central bank actually faces:

- Float and target inflation $\Rightarrow$ exchange rate absorbs FX/commodity-price shocks, but at the cost of nominal volatility and possibly balance-sheet risk.
- Peg $\Rightarrow$ eliminates FX risk, but forces *all* adjustment onto domestic prices and output — amplified by whichever sectors are most import-exposed and most central in the network.
- Manage the float $\Rightarrow$ partially, but at what welfare cost relative to the theoretical optimum?

Rubbo's closed-economy answer (target the DC index) says nothing about *which* exchange-rate regime to combine it with, or how network position shapes the answer. That is the gap this paper fills.

Headline Result: Managed Float Wins the Regime Comparison

Jump to the answer first, mechanism after: solving the full nonlinear open-economy network model for all three FX regimes gives a clear — and non-obvious — welfare ranking.



Managed float dominates both corner regimes — less than half the loss of free float, and lower than a hard peg too. This is the model's own solution to the FX-regime choice, not an assumption.

Why this is surprising: float directly targets the divine-coincidence index and should, on the textbook intuition, be (weakly) best. It is not — floating turns the exchange rate into a slow, persistent state variable that keeps feeding cost-push shocks back into the network (mechanism in the Welfare section).

Roadmap: Literature → This paper → Model → Theory (why DC targeting could fail to survive) → Calibration → Shocks/IRFs → network amplification → welfare cost → shock decomposition.

Literature

- **La'O & Tahbaz-Salehi (2022, *Econometrica*)** — optimal monetary policy in production networks; efficient allocations require targeting a Domar-weighted index, not simple CPI. Closed economy.
- **Rubbo (2024, *Econometrica*)** — the paper we extend. Sectoral Calvo pricing + IO network ⇒ unique DC index with *no* cost-push residual; welfare-optimal Taylor rule targets it.
- **Auer, Levchenko & Sauré (2019, *ReStud*)** — international inflation spillovers propagate through global IO linkages; empirical evidence that network position matters for cross-border pass-through.
- **Pasten, Schoenle & Weber (2020, *JPE*)** — heterogeneous price rigidity across sectors reshapes monetary transmission; motivates letting $\hat{\delta}_i$ differ by sector here too.
- **Fanelli & Straub (2021, *JPE*)** — theory of sterilized FX intervention as a second instrument alongside the policy rate.
- **Gopinath & Itskhoki (2022)** — dominant currency pricing; **Amiti, Itskhoki & Konings (2019, *QJE*)** — variable markups shape exchange-rate pass-through at the firm level.

Gap: no existing paper combines a multi-sector Calvo network *and* a genuine open-economy block (UIP, imported inputs, FX regimes) to ask which FX regime is optimal, and whether the DC-targeting result survives.

This Paper: Contribution and Approach

What we do:

- Extend Rubbo's network Phillips curve to a small open economy: imported intermediate inputs (with an import-cost-share matrix Ω^F), imported final consumption, UIP with a debt-elastic premium.
- Ask which FX regime (float / peg / managed float) is welfare-optimal in a multi-sector network, and whether the DC index survives when the exchange rate is a second cost-push channel.
- Calibrate to a 3-sector oil-exporting economy (Resource $\to$ Manufacturing $\to$ Services).

Methodological choice: the model is built and solved as the *original nonlinear* system — Cobb-Douglas cost minimization, the exact recursive Calvo pricing problem, CRRA/Frisch household FOCs, CES consumption aggregation — rather than substituting in a hand-linearized Phillips curve. The nonlinear steady state is computed numerically and the system perturbed around it, so the log-linear coefficients (e.g. $\hat{\delta}_i$) come out of the solution rather than being assumed.

Notation: Households, Firms, and the Network

Households

C_t, L_t	aggregate consumption, labor supply
β, γ, φ	discount factor; risk aversion (1/IES); inv. Frisch
W_t, i_t	nominal wage; domestic nominal rate
ω, η	home-bias weight; home/foreign elasticity
β_i^H	sector-i share of the domestic consumption bundle
C_t^H, C_t^F	domestic-bundle, imported-bundle consumption
B_t^*	real holdings of the internationally traded bond

Network objects

Ω_{ij}^H	sector i's cost share of domestic input j
Ω_i^F	sector i's cost share of the imported input
$\lambda_{D,i}$	Domar weight (sector i's domestic sales centrality)
$\mathcal{M}_i$	import centrality of sector i
w_i^{DC}	sector i's weight in the DC index

Firms, sector i

Y_{it}, A_{it}	gross output; total factor productivity
L_{it}	labor employed in sector i
X_{ijt}	sector i's use of sector j's output as input
M_{it}	sector i's imported intermediate input
α_i	sector i's labor cost share ($\alpha_i + \sum_j \Omega_{ij}^H + \Omega_i^F = 1$)
P_{it}, MC_{it}	sector-i price; nominal marginal cost
ε	within-sector elasticity (markup $= \varepsilon/(\varepsilon - 1)$)
δ_i	Calvo reset probability (primitive)
$\hat{\delta}_i$	log-linear "effective slope" (derived, not primitive)
$P_{it}^{\#}$	sector-i reset price

Price indices

P_t^H	price of the domestic-goods bundle
P_t^C	CPI (dual price index to C_t)

Notation: Open Economy and Policy

Open economy

S_t	nominal exchange rate (home ccy. per unit foreign ccy.)
P_t^{F*}	foreign-currency price of imports
$P_t^{FH} = S_t P_t^{F*}$	landed domestic-currency import price
i^*	foreign nominal interest rate
ψ	debt-elastic risk-premium coefficient
$\hat{b}_t^*$	(scaled) net foreign asset position
EX_t, IM_t	real exports, real imports
θ^*	export price (demand) elasticity
D_t^*	foreign demand shifter

Policy and shocks

π_{it}, π_t^C	sector-i inflation; CPI inflation
π_t^{DC}	divine-coincidence index inflation
$\tilde{y}_t$	output gap
ϕ_π, ϕ_y, ϕ_s	Taylor-rule weights (inflation, gap, FX)
$\bar{S}$	peg target for the exchange rate
r_t^{nat}	natural real interest rate
a_{it}	sector-i TFP shock, AR(1)
p_t^{F*}, D_t^*	import-price, foreign-demand shocks, AR(1)

All variables denote log-deviations from steady state in the linearized formulas (Theory section); the model-setup slides that follow use the exact nonlinear level variables (capital letters), which is what is actually solved and simulated.

Model Setup: Households — Preferences and Budget

Preferences (CRRA consumption, Frisch labor supply; King, Plosser & Rebelo 1988):

$$\max \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

Sequential budget constraint (representative household; Π_t = firm profits rebated lump-sum; B_t^* = real holdings of the internationally traded bond, valued at S_t). The household chooses $\{C_t, L_t, B_t^*\}_{t\geq 0}$ subject to this each period:

$$P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*)\left[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)\right] S_t B_{t-1}^*$$

No domestic bond needed: representative household, net-zero supply $\Rightarrow$ redundant asset (Wallace irrelevance); the internationally traded bond alone prices consumption smoothing.

Consumption aggregation: CES nest between a domestic bundle and an imported bundle, home-bias ω, elasticity η (Armington 1969; Galí & Monacelli 2005); the domestic bundle is Cobb-Douglas across sectors with shares β_i^H:

$$C_t = \left[\omega^{1/\eta}(C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta}(C_t^F)^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i \left(\frac{C_{it}}{\beta_i^H}\right)^{\beta_i^H}$$

Dual price indices (cost of one unit of C_t^H and C_t respectively; P_t^{FH} defined on the Open-Economy slide):

$$P_t^H = \prod_i P_{it}^{\beta_i^H}, \qquad P_t^C = \left[\omega(P_t^H)^{1-\eta} + (1-\omega)(P_t^{FH})^{1-\eta}\right]^{\frac{1}{1-\eta}}$$

Model Setup: Households — Optimality Conditions

FOCs (from the Lagrangian on the budget constraint, choice variables C_t, L_t, B_t^*; exact nonlinear conditions):

Euler equation ($\partial/\partial C_t$, combined with $\partial/\partial C_{t+1}$):

$$C_t^{-\gamma} = \beta(1+i_t)\,\mathbb{E}_t\left[C_{t+1}^{-\gamma}/\Pi_{t+1}^C\right]$$

Labor supply ($\partial/\partial L_t$):

$$W_t/P_t^C = C_t^\gamma L_t^\varphi$$

Foreign-bond FOC ($\partial/\partial B_t^*$):

$$C_t^{-\gamma} = \beta(1+i^*)\left[1-\psi(\hat{b}_t^*-\bar{b}^*)\right]\mathbb{E}_t\left[C_{t+1}^{-\gamma}\,\frac{S_{t+1}}{S_t\Pi_{t+1}^C}\right]$$

Where UIP comes from: eliminating the common term $\mathbb{E}_t\left[C_{t+1}^{-\gamma}/\Pi_{t+1}^C\right]$ between the Euler equation and the foreign-bond FOC above gives exactly the uncovered interest parity condition used throughout the open-economy block — it is *derived* from household optimization, not assumed:

$$(1+i_t) = (1+i^*)\left[1-\psi(\hat{b}_t^*-\bar{b}^*)\right]\mathbb{E}_t\left[S_{t+1}/S_t\right]$$

Model Setup: Firms — Technology and Cost Minimization

Production (Cobb-Douglas, sector i; network structure as in La'O & Tahbaz-Salehi 2022, Rubbo 2024):

$$Y_{it} = A_{it}\, L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H}\, M_{it}^{\Omega_i^F}$$

Cost minimization: choose $\{L_{it}, X_{ijt}, M_{it}\}$ to minimize total cost $W_t L_{it} + \sum_j P_{jt} X_{ijt} + S_t P_t^{F*} M_{it}$ subject to the production constraint above. The FOCs equate each input's expenditure to its Cobb-Douglas exponent times total cost $MC_{it} Y_{it}$:

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \qquad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \qquad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Summing the three input demands back into the cost function gives nominal marginal cost, multiplicatively separable in input prices:

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

This is the object that both the exchange rate (via M_{it}, Open-Economy slide) and TFP shocks A_{it} shift — the two cost-push channels the rest of the paper is about.

Model Setup: Firms — Calvo Price Setting

Calvo pricing (Calvo 1983): with probability δ_i (primitive; *not* $\hat{\delta}_i$, derived below) the firm resets its price each period; with probability $1-\delta_i$ it keeps last period's price. A resetting firm chooses $P^{\#}_{it}$ to maximize expected discounted profits, weighting each future state by the probability $(1-\delta_i)^s$ that the price is *still* not reset s periods later:

$$\max_{P^{\#}_{it}} \ \mathbb{E}_t \sum_{s=0}^{\infty} \rho^s (1-\delta_i)^s \left[P^{\#}_{it} - MC_{it+s}\right] \left(\frac{P^{\#}_{it}}{P_{it+s}}\right)^{-\varepsilon} Y_{it+s}$$

where $\left(P^{\#}_{it}/P_{it+s}\right)^{-\varepsilon} Y_{it+s}$ is the CES demand curve the firm faces (elasticity ε, markup $\varepsilon/(\varepsilon-1)$). FOC (forward-looking markup pricing):

$$P^{\#}_{it} = \frac{\varepsilon}{\varepsilon-1} \cdot \frac{\mathbb{E}_t \sum_s \rho^s (1-\delta_i)^s Y_{it+s} \widetilde{MC}_{it+s}}{\mathbb{E}_t \sum_s \rho^s (1-\delta_i)^s Y_{it+s}/P_{it+s}}$$

Reading the FOC: the reset price is a markup over a *weighted average of current and future marginal costs*, with weights given by how likely the price is to still be in effect. The stickier the sector (δ_i small), the further forward the firm must look — this forward-lookingness is exactly what the recursive form on the next slide computes period by period.

Model Setup: Firms — Calvo Recursion, as Solved

As coded (recursive form of the FOC on the previous slide, real MC $\widetilde{MC} = MC/P$): $X1_{it}$ and $X2_{it}$ are the numerator and denominator of the reset-price formula, written recursively so Dynare can solve them jointly with the rest of the nonlinear system:

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\Pi^{\varepsilon}_{i,t+1}X1_{i,t+1}$$

$$X2_{it} = Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\Pi^{\varepsilon-1}_{i,t+1}X2_{i,t+1}, \qquad P^{\#}_{it}/P_{it} = \frac{\varepsilon}{\varepsilon-1}X1_{it}/X2_{it}$$

Price index law of motion (CES aggregation of the δ_i fraction that resets to $P^{\#}_{it}$ and the $1-\delta_i$ fraction that keeps last period's price):

$$1 = (1-\delta_i)\Pi^{\varepsilon-1}_{it} + \delta_i\left(P^{\#}_{it}/P_{it}\right)^{1-\varepsilon}$$

Why this matters: log-linearizing this exact recursion around the steady state is what *produces* Rubbo's adjusted slope $\hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$, which appears throughout the Theory section — it is a derived object, not a primitive of the model.

Model Setup: Open Economy — Prices and Interest Parity

Law of one price for imports (S_t: nominal ER, home ccy. per unit foreign ccy.; P_t^{F*}: foreign-currency import price; Galí & Monacelli 2005):

$$P_t^{FH} = S_t P_t^{F*}$$

This is the last piece needed to close the price indices from the Households slide: $P_t^H = \prod_i P_{it}^{\beta_i^H}$ is the price of the domestic bundle, and $P_t^C = \left[\omega(P_t^H)^{1-\eta} + (1-\omega)(P_t^{FH})^{1-\eta}\right]^{1/(1-\eta)}$ is the CPI.

Uncovered interest parity, with a debt-elastic risk premium (functional form: Schmitt-Grohé & Uribe 2003, for stationarity of $\hat{b}_t^*$). Derived, not assumed: eliminate $\mathbb{E}_t C_{t+1}^{-\gamma}/\Pi_{t+1}^C$ between the household's domestic Euler equation and its foreign-bond FOC (Households slide) to get

$$(1 + i_t) = (1 + i^*)\left[1 - \psi(\hat{b}_t^* - \bar{b}^*)\right] \mathbb{E}_t \left[S_{t+1}/S_t\right]$$

Under a float, i_t follows the Taylor rule and S_t adjusts to satisfy this equation; under a peg, $S_t = \bar{S}$ is fixed and i_t is the residual.

Key mechanism: imported inputs enter every sector's marginal cost through Ω_i^F (Firms slide, via M_{it}) — the exchange rate now moves marginal cost *exactly like a productivity shock*, scaled by each sector's import exposure and propagated through Ω^H.

Model Setup: Open Economy — External Accounts and Market Clearing

Net foreign assets (aggregating household budget constraints across the economy collapses to the standard current-account law of motion):

$$\hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Trade flows: EX_t is reduced-form foreign demand for the home bundle (elasticity θ^*, shifter D_t^*); IM_t sums final-consumption imports and the imported input embedded in every sector's cost:

$$EX_t = \kappa_{EX}\, D_t^* \left(\frac{P_t^H}{P_t^{FH}}\right)^{-\theta^*}, \qquad IM_t = C_t^F + \sum_i \Omega_i^F \frac{MC_{it} Y_{it}}{P_t^{FH}}$$

Sectoral goods-market clearing: sector i's output is absorbed by domestic consumption and exports of i's share of the bundle, plus intermediate demand from every downstream sector j that uses i as an input:

$$Y_{it} = \beta_i^H \frac{P_t^H}{P_{it}}\left(C_t^H + EX_t\right) + \sum_j \Omega_{ji}^H \frac{MC_{jt} Y_{jt}}{P_{it}}$$

(In the 3-sector calibration: Resource feeds Manufacturing, Manufacturing feeds Services, and each sector also sells directly to C_t^H and EX_t.)

Policy Regimes and Shocks

Three FX regimes (only the policy block changes):

Regime	Rule	Exchange rate
Float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual from UIP
Managed float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Three shocks studied:

- Sectoral TFP shocks a_{it}, AR(1), $\rho_a = 0.90$
- Import price shock p_t^{F*}, AR(1), $\rho_{pF} = 0.85$ — the pure FX/commodity-price channel
- Foreign demand shock D_t^*, AR(1), $\rho_D = 0.80$ — affects exports, hence NFA and the exchange rate

Rubbo (2024): What Carries Over

Closed-economy system (N sectors, Calvo $\hat{\delta}_i$, IO network Ω):

$$\begin{aligned} \text{IS:}\quad & \tilde{y}_t = \mathbb{E}\tilde{y}_{t+1} - \tfrac{1}{\gamma}(i_t - \mathbb{E}\pi^C_{t+1} - r^{nat}_t) \\ \text{MC:}\quad & \boldsymbol{mc}_t = \boldsymbol{\alpha} w_t + \Omega \boldsymbol{p}_t - \log \mathbf{A}_t \\ \text{NKPC:}\quad & \boldsymbol{\pi}_t = \rho(I-\mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma+\varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t \\ \text{Policy:}\quad & i_t = r^{nat}_t + \phi_\pi \pi^C_t + \phi_y \tilde{y}_t \end{aligned}$$

Lemma 1: natural output = Domar-weighted TFP, $y^{nat}_t = \sum_i \lambda_i a_{it}$.
Proposition 1 (DC index): unique index with *no* cost-push:

$$\pi^{DC}_t = \rho\,\mathbb{E}\pi^{DC}_{t+1} + \frac{\gamma+\varphi}{\Lambda^{DC}}\tilde{y}_t, \quad w^{DC}_i \propto \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Why DC is unique: $\mathcal{V} = \Delta(I-\Omega\Delta)^{-1}$ has rank $N-1$ on the space of cost-push shocks $\Rightarrow$ a one-dimensional null space $\Rightarrow$ exactly one index ϕ with $\phi^\top\mathcal{V} = \mathbf{0}^\top$.

One shock, one channel: in the closed economy, only TFP shocks $\boldsymbol{\chi}_t$ enter the cost-push term. That is exactly why a single index can null it out.

The open question we ask: with a *second* cost-push channel (the exchange rate), can the same trick still null out both simultaneously?

Why the Divine Coincidence Matters

Where the name comes from (Blanchard & Galí 2007, *JEEA*): in the canonical one-sector NK model, if the *only* distortion is nominal price stickiness, then stabilizing inflation, closing the output gap, and maximizing welfare are the *same* objective — one instrument, one target, no tradeoff. It is called "divine" because this is fragile: any second distortion (real wage rigidity, markup or cost-push shocks) breaks it and forces a genuine tradeoff.

Why sectoral heterogeneity threatens it: with N sectors resetting prices at different rates δ_i, an arbitrary index — plain CPI, say — behaves like a model with an extra distortion: stabilizing it no longer stabilizes the output gap. Rubbo's Proposition 1 shows the coincidence is not lost, only *hidden*: exactly one Domar- and stickiness-weighted index restores it (previous slide).

Why it matters in practice:

- It gives a central bank a single, well-defined, model-consistent target whose stabilization *alone* delivers the fully efficient (flexible-price) allocation — no discretionary weighing of inflation against output.
- It shows plain CPI targeting is *not* optimal once sectors differ in stickiness: the wedge between CPI inflation and π_t^{DC} is exactly the welfare-relevant cost-push term a CPI-targeting central bank would be missing.
- It is not merely a price index: by Proposition 3, $\pi_t^{DC} \equiv 0$ for all t exactly minimizes the model's quadratic welfare loss — "divine coincidence" and "welfare-optimal policy" are the same statement.

Why it recurs on almost every slide that follows: the paper is organized around one question — does this single tradeoff-free target survive once the exchange rate becomes a *second* source of cost-push shocks. Every quantitative result below is, implicitly, a test of how "divine" the coincidence remains once the economy is open.

Open-Economy Divine Coincidence: Does It Survive?

Closed economy (Rubbo): one cost-push channel (TFP) $\Rightarrow$ DC index exists uniquely because a rank-deficient matrix $\mathcal{V}$ has a one-dimensional null space, and one index can sit in it.

Open economy: two cost-push channels. TFP *and* the exchange rate now shift sectoral marginal cost. A single weight vector would need to null out *both* simultaneously:

$$\underbrace{\phi^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP channel}} \quad \text{and} \quad \underbrace{\phi^\top \Gamma = 0}_{\text{FX channel}}$$

Two conditions, one free vector $\Rightarrow$ generically no solution: **DC breaks down under a float.**

But: import nodes can be modeled as fully flexible upstream sectors ($\hat{\delta}_M = 1$) in an augmented network. Fully flexible nodes get *zero* DC weight ($w_M = 0$ since $(1 - \hat{\delta}_M)/\hat{\delta}_M = 0$), so Rubbo's Proposition 1 still applies to the *augmented* network — the DC index (as coded here) remains well-defined and insulates against domestic TFP shocks by construction. Whether it *also* nulls the FX channel in the calibrated network is the quantitative question tackled below.

Regime Comparison: Theoretical Predictions

	Float	**Peg**	**Managed**
Exchange rate	fully endogenous	fixed	partially stabilized
FX cost-push absorbed by	S_t adjustment	domestic π_i, y	split
Monetary independence	full	none (residual i_t)	full
Pass-through of import shocks	low (if ϕ_π high)	proportional to $\mathcal{M}_i$	intermediate
DC index well defined?	yes (augmented network)	n/a (S_t not a choice)	yes

Naive expectation: float should dominate on inflation stabilization since it directly targets π_t^{DC}; peg transmits import shocks directly, scaled by $\mathcal{M}_i$; managed sits in between.

Preview: the quantitative results overturn part of this. Solving the actual nonlinear model shows free float is *not* the best regime here — see the Welfare section, below.

Solution Method: Nonlinear, Not Pre-Linearized

Why solve the nonlinear model directly, rather than hand-linearize first:

- The Calvo slope $\hat{\delta}_i$ and the DC weights are *outputs* of log-linearizing the true recursion — coding them as inputs risks baking in the answer.
- A genuine nonlinear model lets us go to order-2 perturbation, which is required for valid welfare comparisons across policy regimes (Kim-Kim / Woodford: first-order-accurate welfare rankings are not valid — variance terms enter the welfare function only at second order).

Steady state: solved semi-analytically, not guessed. At zero inflation, price = markup × marginal cost exactly, which collapses the 46-equation system to one scalar root-find in the real wage W (prices, exports, consumption all closed-form given W), giving $W^* = 1.3189$.

- **Lemma 1 (exact):** $\tilde{y}_t = \sum_i \lambda_{D,i} \left[\log(Y_{it}/\bar{Y}_i) - \log A_{it} \right]$ — no shadow flex-price block needed.
- **Prop. 1 weights (theory-motivated):** DC weights from $\lambda_{D,i}$, $\hat{\delta}_i$, applied to realized inflation in the true nonlinear simulation.

Solved at first order around the exact steady state, for all three regimes. Second-order perturbation is not used: an inflation/output-gap-only Taylor rule leaves the price *level* and S_t as a unit root — expected, since nothing anchors the level — so only the first-order (stationary) solution is used, and only stationary objects enter welfare.

Calibration: A 3-Sector Oil-Exporting Economy

Parameter	**Value**	**Parameter**	**Value**
β (discount factor)	0.99	Ω^H_{21} (Manuf. $\leftarrow$ Resource)	0.20
γ (risk aversion)	1.00	Ω^H_{32} (Services $\leftarrow$ Manuf.)	0.25
φ (inv. Frisch elasticity)	2.00	$\Omega^F_1, \Omega^F_2, \Omega^F_3$ (import shares)	0.30, 0.10, 0.05
η (home/foreign elasticity)	1.50	$\beta^H_1, \beta^H_2, \beta^H_3$ (consumption shares)	0.05, 0.15, 0.80
$\beta^{F,tot}$ (import share, CPI)	0.10	$\delta_1, \delta_2, \delta_3$ (Calvo reset prob.)	0.75, 0.50, 0.25
ε (within-sector elasticity)	8.00	ψ (NFA debt-elastic premium)	0.020
θ^* (export price elasticity)	2.00	ϕ_π, ϕ_y, ϕ_s (policy)	1.50, 0.50, 0.30

Sector story: Resource is upstream, flexible-priced, import-intensive (machinery) and the main exporter. Manufacturing is the middle of the chain. Services is downstream, sticky-priced, and dominates final consumption — the classic setting for network amplification.

Network Properties: Centrality and DC Weights

Domestic supplier centrality (Domar weight):

$$\lambda_{D,i} = \left[(\beta^H)^\top (I - \Omega^H)^{-1}\right]_i$$

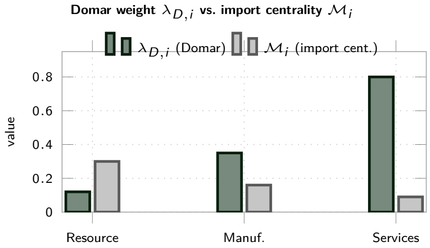
Import centrality (FX exposure transmitted through the domestic network):

$$\mathcal{M}_i = \left[(I - \Omega^H)^{-1} \Omega^F \mathbf{1}\right]_i$$

DC-index weights (Proposition 1, applied here to the true nonlinear realized inflation):

$$w_i^{DC} \propto \lambda_{D,i} \frac{1 - \hat{\delta}_i}{\hat{\delta}_i}$$

Both objects are computed once from the calibrated $\Omega^H, \Omega^F, \beta^H$ matrices — no magic numbers, same formulas as Rubbo's Proposition 1 and the import-centrality analog.



Expect: Services has the largest $\lambda_{D,i}$ (dominates final demand, downstream); Resource has the largest $\mathcal{M}_i$ (directly import-intensive) but this need not be the sector with the largest DC weight, since DC weights also load on stickiness $(1 - \hat{\delta}_i)/\hat{\delta}_i$.

From Theory to Quantification

Pipeline: the model is solved numerically for all three regimes, producing IRFs and theoretical moments, summarized in the figures below. Every number on the following slides is model output for the calibration two slides ago.

Main text focuses on the import-price shock — the cleanest test of the FX cost-push channel the Theory section just introduced, since it hits every sector's marginal cost through Ω_i^F with no domestic-TFP confound. (TFP and foreign-demand IRFs, which confirm the same ranking, are in the Appendix.)

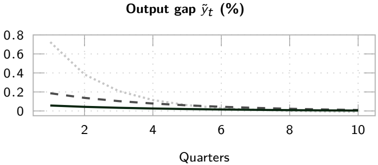
Five questions the simulations answer:

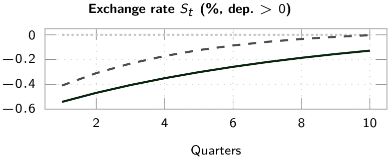
1. Which sectors are most exposed to the import-price shock, and does that line up with Domar weight, import centrality, or both?
2. Does the DC index stay close to zero (insulation property), or does it visibly move under the float?
3. What is the welfare ranking of float vs. peg vs. managed float, and what drives it – output-gap volatility or sector-level price dispersion?
4. Is the driver on-impact volatility, or something slower (persistence)?
5. How much of this depends on *this particular* calibration – i.e. does the ranking generalize? (Generalization section, later in the deck.)

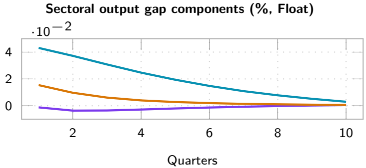
Headline (surprising) answer: free float is *not* the best regime here. Managed float dominates on almost every

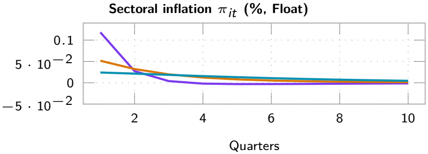
Results: IRFs to an Import-Price Shock — the Mechanism Shock

A 1% increase in the foreign-currency import price p_t^{F*}, $\rho_{pF} = 0.85$ — the shock that directly targets Γ, the FX cost-push channel from the Theory section.



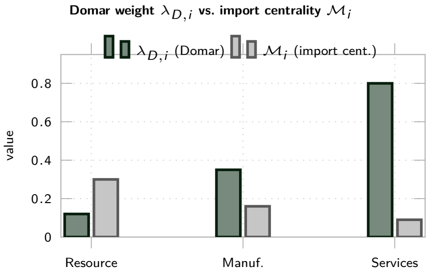


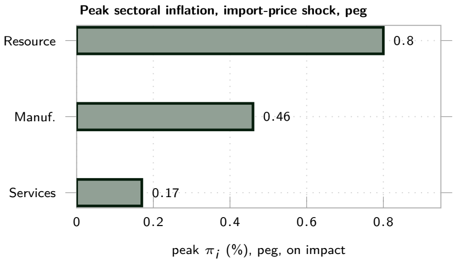




Top row (Peg / Managed / Float): ▪ ▪ ▪ Peg - - Managed — Float. Bottom row (Float regime, by sector): — Resource

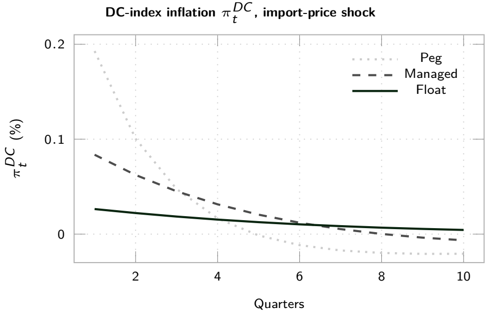
Results: Sectoral Heterogeneity and Network Centrality





Two different rankings, confirmed by simulation: Resource has the highest import centrality $\mathcal{M}_i = 0.30$ and (as predicted) the most volatile own inflation on impact (0.80%), but the lowest Domar weight $\lambda_{D,i} = 0.12$ — its shocks barely move the aggregate. Services is the mirror image: low import exposure ($\mathcal{M}_3 = 0.09$), the least volatile own inflation on impact (0.17%), but dominates the aggregate (Domar weight 0.80, DC weight 0.93). As the welfare numbers below show, "least volatile on impact" does *not* mean "least costly" once persistence is priced in.

Results: DC-Index Insulation and Trade Flows



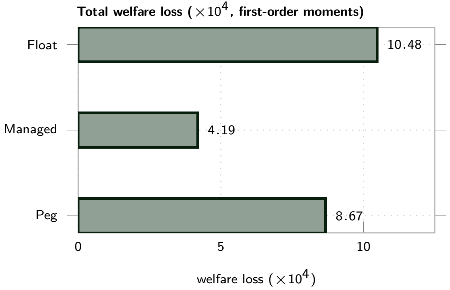
Peak response to a foreign-demand shock

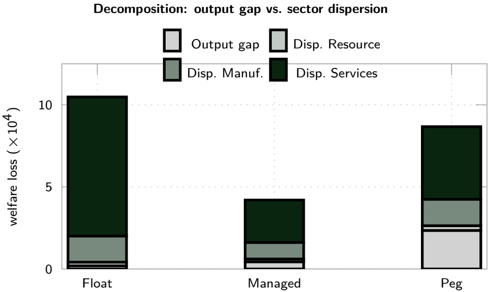
(%):	Float	Managed	Peg
Exports EX_t	0.122	0.134	0.182
Imports IM_t	0.081	0.090	0.130
NFA $\hat{b}_t^*$	0.122	0.122	0.119
Output gap	0.064	0.115	0.339

NFA accumulation is almost identical across regimes; the output gap is not — peg's output gap moves $\sim 5\times$ more than float's for the same demand shock, since the interest rate cannot help absorb it.

DC insulation, on impact: float's π_t^{DC} is smallest on impact (0.026%) — the augmented-network DC index does dampen the immediate FX cost-push. But float's response is also the *most persistent* (barely decaying by quarter 10), while peg's and managed's both overshoot and reverse sign. On-impact insulation is not the whole story.

Results: Welfare Cost by Regime — The Headline Finding

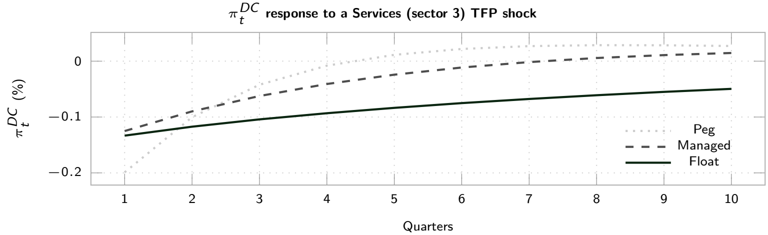




Welfare measure (Rubbo Prop. 3, adapted): $\mathcal{W} = \frac{1}{2}\left[(\gamma+\varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i}\varepsilon_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\mathrm{Var}(\pi_{it})\right]$, evaluated on Dynare's own order-1 theoretical variances (see caveat on order vs. welfare, Calibration section). **Float has the lowest output-gap cost of all three (0.19) but by far the largest Services-dispersion cost (8.48, driven by the persistence channel below) — managed float is the only regime that is not dominated on some margin.**

Why Floating Costs More Than Expected: Persistence, Not Volatility

Mechanism: under float, S_t is pinned down by UIP and the (highly persistent, autocorrelation 0.994) net-foreign-asset process $\hat{b}_t^*$. Since S_t enters *every* sector's marginal cost via Ω_i^F, letting the currency float doesn't just insulate against FX shocks — it also lets *domestic* TFP shocks propagate through a new, slow-decaying exchange-rate channel that does not exist under a peg ($S_t \equiv 0$).



Reading: peg resolves the shock via a fast, one-time adjustment and overshoots/reverses within 3–4 quarters; float's response barely decays over the whole horizon. This is a *pure domestic TFP shock* — no FX shock to insulate against — yet float still has the most drawn-out response, since the exchange rate is now a slow state variable feeding continuously back into costs.

Which Sector Prefers Which Regime? Two Equations

Two calibratable objects, not one, determine a sector's regime preference — and in this economy they point in *opposite* directions across sectors.

1. Direct FX exposure (network- and rigidity-weighted import share):

$$\Gamma_i = \frac{[\Delta(I - \Omega\Delta)^{-1}\omega_F]_i}{\kappa}$$

	Resource	Manuf.	Services
Γ_i (FX exposure)	**0.234**	0.054	0.006
κ_i (welfare weight)	0.43	5.54	**74.6**

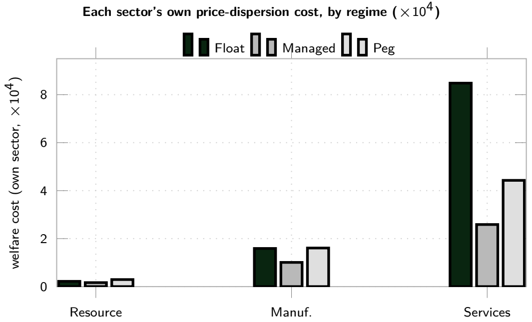
2. Welfare weight (centrality × stickiness):

$$\kappa_i = \lambda_{D,i}\,\varepsilon\,\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

The sector most exposed to the exchange rate on impact is the one welfare cares about least; the sector welfare cares about most is almost insulated from it directly. A peg switches off $\Gamma_i \Delta e_t$ for everyone — but it does not eliminate adjustment, it *relocates* it to domestic prices, landing on the upstream sector because it sits at the border (P^H is dominated by its price).

Net effect (real $\kappa_i \mathrm{Var}^R(\pi_i)$, $\times 10^4$): Resource's relocated-adjustment cost under peg outweighs its avoided Γ-cost (0.44 → 0.59) — so despite the largest Γ_i, it still prefers *float*. Services is insulated from both channels, so peg nearly halves its inflation variance (16.96 → 8.86) — and since $\kappa_3 \gg \kappa_1, \kappa_2$, this one sector's preference dominates the aggregate ranking.

Sector-Level Winners and Losers



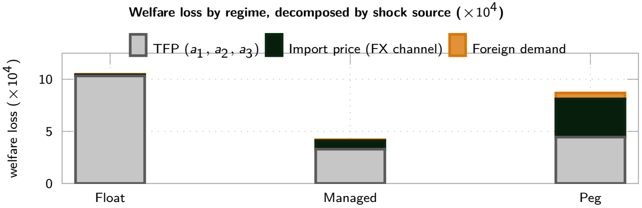
All three sectors agree on their first choice: managed float. They disagree on the runner-up:

- **Resource** (flexible, peripheral, most import-exposed): Managed $>$ Float $>$ Peg. Direct FX exposure makes the peg its worst option.
- **Manufacturing**: Managed $\gg$ $\{$Float $\approx$ Peg$\}$ — roughly indifferent between the two corner regimes.
- **Services** (sticky, central, low import exposure): Managed $>$ Peg $>$ Float. The persistence channel (previous slide) hurts it more than direct FX exposure does, so it would rather fix the exchange rate than float it.

Key point: the sector that is most exposed to FX shocks directly (Resource) and the sector with the most at stake in the regime choice overall (Services) are not the same sector, and they even rank the two corner regimes in *opposite* order — a policymaker averaging across sectors could easily miss this.

Which Shock Actually Drives the Welfare Loss?

The Welfare section pools all five shocks into one number. Dynare's own `variance_decomposition` (%) splits each regime's total welfare loss by shock *source*, with no re-estimation needed.



The FX-vs-TFP intuition gets it backwards for float and managed: in *every* regime, TFP shocks – not FX shocks – are the dominant source of welfare loss (10.32 of Float's 10.48; 3.29 of Managed's 4.19). **Peg is the only regime where the FX channel is a first-order contributor** (3.66, ≈42% of its 8.67 total) – it has no instrument left to absorb it. **Float's FX-channel cost is nearly zero** (0.11, S_t genuinely insulates) **but that same floating S_t is what makes TFP shocks so costly** (persistence mechanism, Welfare section): regime choice looks like it's about FX exposure but is mostly about TFP-shock persistence.

Note: this decomposition currently spans TFP, import-price, and foreign-demand shocks. The newly added

From "An Oil Exporter" to "Which Structural Features Matter"

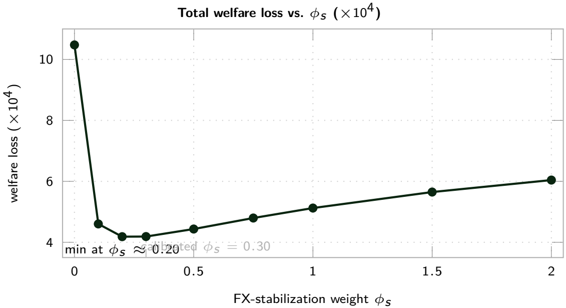
The calibration above is one economy. The Theory section's result — the DC index breaks down under a float because $\phi^\top \Gamma = 0$ and $\phi^\top \mathcal{V} = \mathbf{0}^\top$ generically cannot hold simultaneously — is a statement about *any* network with $\Gamma \not\propto \mathcal{B}$ (FX exposure not proportional to labor intensity). This section asks which structural features govern *how costly* that breakdown is, using three re-solves of the full nonlinear model (not new shocks – the same three regimes, the same welfare metric):

1. **Policy-space sensitivity (ϕ_s):** for a fixed economy, how much FX stabilization is optimal? (already-calibrated managed-float rule, re-solved on a grid.)
2. **Import-intensity level:** scale the whole import-cost-share vector Ω^F up/down, holding its *relative* sectoral pattern fixed. Tests whether the peg-can-beat-float result is an artifact of this economy's degree of import dependence.
3. **Import-exposure heterogeneity:** hold *total* import intensity fixed but vary how unevenly it is spread across sectors (Resource/Manuf./Services). Tests the theorem directly: less heterogeneity should move Γ closer to proportional, and the float/peg welfare gap should shrink.

Reading ahead: if the managed-float result is fragile to (2)–(3), it is a fact about oil exporters; if it survives, it is a fact about production networks with heterogeneous FX exposure – a much stronger claim.

Sensitivity: The Optimal Degree of FX Stabilization

Sensitivity sweep: re-solved the managed-float regime for $\phi_s \in [0, 2]$ ($\phi_s = 0$ is the pure float limit). ϕ_s does not affect the steady state, so this isolates the policy trade-off cleanly.



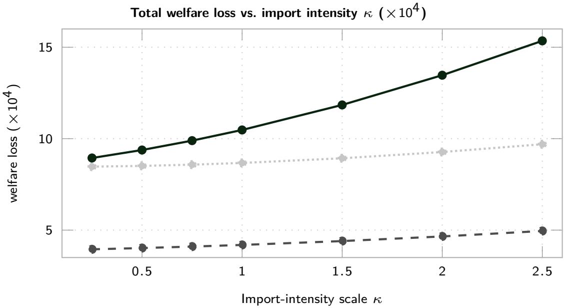
Reading:

- Going from pure float ($\phi_s = 0$) to just $\phi_s = 0.10$ more than halves the welfare loss ($10.48 \to 4.60$).
- The minimum is flat and wide: $\phi_s \in [0.2, 0.3]$ is statistically indistinguishable, and the calibrated value (0.30) is very close to optimal.
- Beyond $\phi_s \approx 0.3$, cost rises again — too much FX stabilization crowds out output-gap stabilization from the same single interest-rate instrument.

Bottom line: a little FX stabilization goes a long way; a lot is counterproductive. Neither corner regime (float or peg) is optimal — a modest, positive ϕ_s is.

Robustness: Import-Intensity Level

Sweep 1: scale the whole import-cost-share vector Ω^F by $\kappa \in [0.25, 2.5]$ ($\kappa = 1$ is the baseline oil-exporter calibration), holding its relative sectoral pattern fixed. α_i absorbs the change so Cobb-Douglas shares still sum to 1.



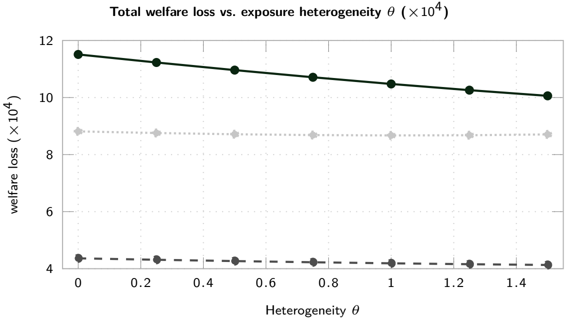
Reading:

- Ranking **managed** $<$ **peg** $<$ **float** holds at *every* κ from 0.25 to 2.5.
- All three losses rise with import dependence, float fastest: gap grows from 0.47 ($\kappa = 0.25$) to 5.65 ($\kappa = 2.5$).
- Managed float is the flattest line — FX stabilization caps the escalating cost of floating.

Bottom line: not an oil-exporter artifact — it strengthens, if anything, as import dependence rises.

Robustness: Import-Exposure Heterogeneity

Sweep 2: hold *total* import intensity fixed but vary θ, how unevenly Ω^F is spread across sectors ($\theta = 0$: identical; $\theta = 1$: baseline; $\theta = 1.5$: more skewed). Tests whether $\Gamma \propto \mathcal{B}$ drives the cost of floating.



Reading:

- Ranking **managed** $<$ **peg** $<$ **float** again holds at every θ.
- **Unexpected:** the float-peg gap *shrinks* as exposure gets *more* uneven (2.70 at $\theta = 0 \to 1.35$ at $\theta = 1.5$) — opposite the naive $\Gamma \propto \mathcal{B}$ prediction.
- Why: α_i absorbs whatever Ω_i^F leaves over given *fixed* Ω^H, so flattening Ω^F disperses α_i instead of aligning Γ with $\mathcal{B}$.

Bottom line: ranking is robust, but this design doesn't cleanly isolate $\Gamma \propto \mathcal{B}$ — a fixed-α_i design is needed for that.

Summary: Key Messages

1. **Free float is not the optimal regime, even though it directly targets the DC index.** Managed float has less than half the total welfare loss of pure float (4.19 vs. 10.48, $\times 10^4$) and beats the hard peg too (8.67). This is the model's own solution, not an assumption.
2. **Why:** floating turns the exchange rate into a slow, persistent state variable (via UIP and the near-unit-root NFA process) that feeds continuously back into every sector's marginal cost through Ω_i^F, turning even domestic TFP shocks into long, drawn-out inflation. **Persistence, not on-impact volatility, drives the welfare cost.**
3. **Every sector prefers managed float first**, but ranks the corner regimes oppositely: Resource (import-exposed, peripheral) ranks float over peg; Services (import-light, central) ranks peg over float.
4. **The optimal FX stabilization is small but strictly positive**: welfare is minimized around $\phi_s \approx 0.2$–0.3, with more than half the gain from just $\phi_s = 0.10$.
5. **The managed $<$ peg $<$ float ranking is robust** to import intensity ($\kappa \in [0.25, 2.5]$) and exposure heterogeneity ($\theta \in [0, 1.5]$) — not an oil-exporter artifact. The heterogeneity gap moves opposite the naive $\Gamma \propto \mathcal{B}$ prediction since this design lets α_i absorb the Ω_i^F change; isolating that mechanism cleanly is future work.
6. These are first-order welfare comparisons (order-2 pruning fails due to the price-level/ER unit root) — rankings are indicative of the mechanism, not certainty-equivalent-exact.

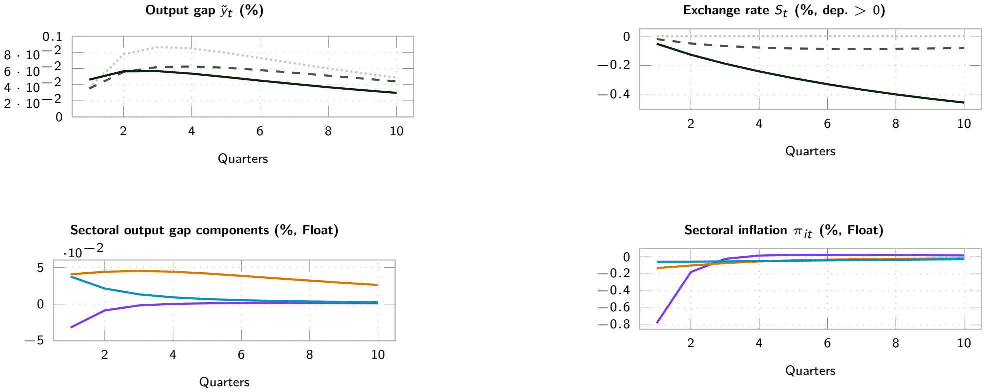
Next Steps

Open questions going forward:

- A price-level-anchored policy rule, to obtain a fully second-order-accurate welfare ranking.
- Formalizing the exchange-rate persistence mechanism analytically (a closed-form counterpart to the Generalization section's numerical result).
- A dedicated world-commodity-price/ToT shock on the export side (`eps_pX`, now wired into the Dynare model) alongside the existing import-price shock – rerun the pipeline and fold the new IRFs and welfare decomposition into the Shocks/Network/Decomposition sections.
- Sensitivity to Calvo stickiness δ_i and to the domestic network shares Ω^H (the import-side sensitivity is now done – see Generalization).

Results: IRFs to a Resource TFP Shock

A 1% Resource-sector (upstream, flexible, import-intensive) TFP improvement, $\rho_a = 0.90$.

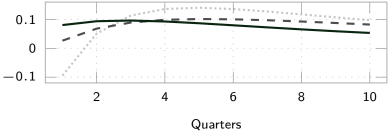


Top row (Peg / Managed / Float): ▪ ▪ ▪ Peg - - Managed — Float. Bottom row (Float regime, by sector): — Resource — Manuf. — Services.

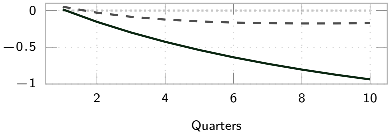
Results: IRFs to a Manufacturing TFP Shock

A 1% Manufacturing-sector (mid-chain) TFP improvement, $\rho_a = 0.90$.

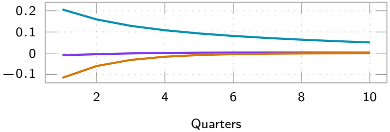
Output gap $\tilde{y}_t$ (%)



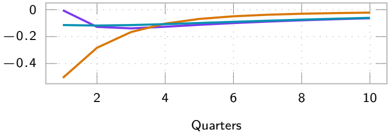
Exchange rate S_t (%, dep. > 0)



Sectoral output gap components (%, Float)



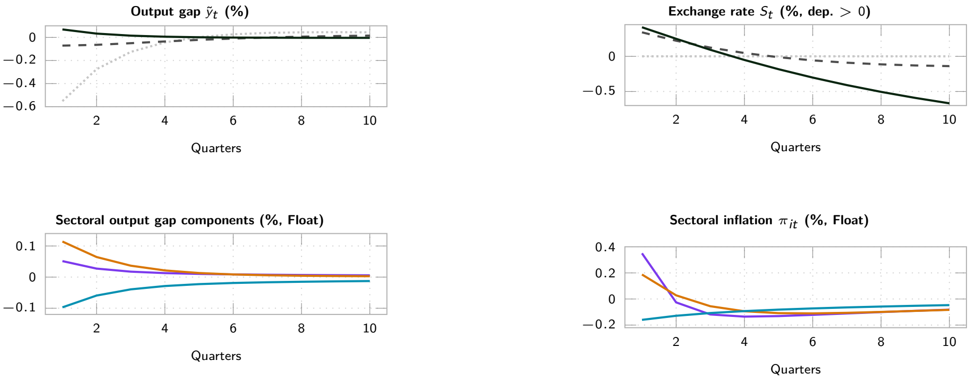
Sectoral inflation π_{it} (%, Float)



Top row (Peg / Managed / Float): ■ ■ ■ Peg - - Managed — Float. Bottom row (Float regime, by sector): — Resource — Manuf. — Services.

Results: IRFs to a Services TFP Shock

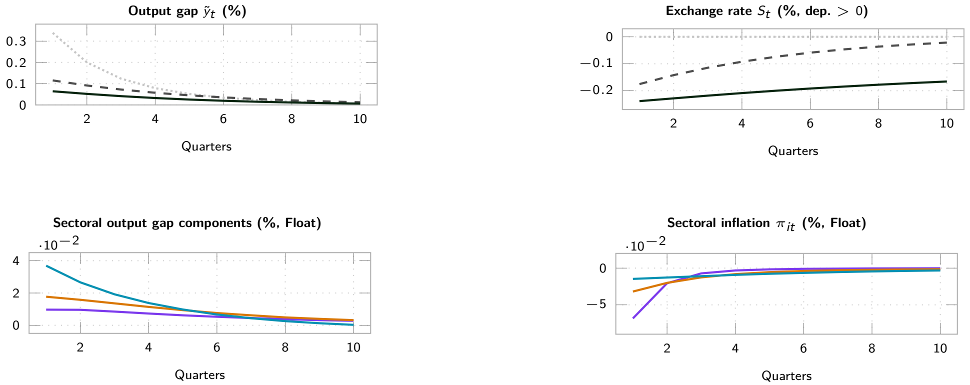
A 1% Services-sector (downstream, sticky, low import exposure) TFP improvement, $\rho_a = 0.90$.



Top row (Peg / Managed / Float): ▪ ▪ ▪ Peg - - Managed — Float. Bottom row (Float regime, by sector): — Resource — Manuf. — Services.

Results: IRFs to a Foreign-Demand Shock

A 1% foreign-demand shock D_t^* (raises export demand for the home bundle), $\rho_D = 0.80$.



Top row (Peg / Managed / Float): ▪ ▪ ▪ Peg - - Managed — Float. Bottom row (Float regime, by sector): — Resource — Manuf. — Services.